

Lab 1 - Radio-Frequency Digital Communication Signals

Goal

In this lab, we will survey several radio-frequency communication signal formats. Our goal is to understand the relationship between the power density spectrum of the transmitted signal, the rate at which symbol pulses are transmitted, and the rate at which bits are transmitted. These signaling formats will be generated by the USRP using software that we will be using throughout the class. We will also measure several “live” communication signals with both the USRP and a stand-alone spectrum analyzer.

1 USRP Generated Signals

1.1 Binary-phase-shift-keyed signals

1. We will start with a digital modulation format that transmits one bit per symbol interval using a single carrier frequency. This format is called binary-phase-shift keying or BPSK. In order to set the modulation rate to BPSK, you will need to find the “PAM” control and enable it (i.e. the green light should be lit).

2. Start *USRP_TX_Digital_AM.vi* and change the following settings:

- a) Carrier Frequency = $(999 + n)M$ where n is the lab station you are working at. So lab station $n = 1$ would use carrier frequency $f_c = 1G$
- b) Device Name: *MyB200*
- c) Antennaa: *TX1*

Note: Make sure to use a **USRP B200** for this part, you will only need **one** radio for Lab1.

3. Start *USRP_RX_Digital_AM.vi* and change the following settings:

- a) Carrier Frequency = $(999 + n)M$ where n is the lab station you are working at. So lab station $n = 1$ would use carrier frequency $f_c = 1G$
- b) Device Name: *MyB200*
- c) Antenna: *RX2*

4. Now, using the newly opened VI, view the following:

- a) The magnitude of the power spectrum of the signal (i.e. the top graph) $\sqrt{I^2 + Q^2}$.
- b) The power spectrum of the in-phase component (i.e. the middle graph) I^2 .
- c) The power spectrum of the quadrature component (i.e. the bottom graph) Q^2 .

Take a screen shot for each of spectra to include in the lab write up.

5. Now vary the phase of the local oscillator (by using the blue slider bar control below the constellation graph) on the *USRP_RX_Digital_AM.vi* and observe what happens to the power in the in-phase component as compared to the power in the quadrature component. Is the magnitude $\sqrt{I^2 + Q^2}$ of the power density spectrum affected by the phase? Why or why not?
6. Is it possible to set the phase so that the power density spectrum for the in-phase component is very close to zero? If the receiver is only sensitive to this component, what would happen?
7. Now move the Tx and Rx antennas relative to one another and see the effect on the power density spectrum. What is the cause of this effect?
8. Again vary the phase of local oscillator on the VI and observe what happens to the signal constellation. Is this behavior consistent with what you observed for the power density spectrum?
9. Set the phase of the local oscillator so that the constellation is aligned along the *I*-axis. Determine the average power of the constellation, which is given by

$$P_{\text{ave}} = \frac{1}{N} \sum_{i=1}^N |s_i|^2 \quad (1)$$

$$|s_i|^2 = s_i (s_i)^*$$

where s_i is the complex value corresponding to the coordinates of each constellation point, and N is the number of constellation points used by the particular modulation format. (This calculation can be done off-line using a screen shot of the constellation.)

10. Now move the antennas relative to one another and see the effect of the motion on the constellation.
11. Reduce the carrier frequency by a factor of two and repeat the same motion. Does the constellation change at the same rate for the lower carrier frequency? Why or why not?
 - a) **Note:** You will need to change the Carrier Frequency for both the Tx and Rx VI's.
12. Offset the carrier frequency between the transmitter and the receiver by 1 Hz and observe the constellation. Why does this happen?
 - a) **Note:** You can dial in a small frequency offset using the “Fine Frequency Offset” control in the Receiver VI.
13. Now move the antennas at a constant rate toward and away from each other. See if you can “freeze” the rate of rotation by moving the antennas. Why does the rate slow down in one direction and increase in the other direction? (Hint: Moving the antennas corresponds producing a small Doppler shift between the transmitter and the receiver.)

Write-Up for Part 1

1. All questions posed in the narrative.
2. All requested screen shots.
3. Write down the symbol rate and the bit rate for each of the two modulation formats used in Part 1. Recall, for a square pulse $p(t)$

$$p(t) = \begin{cases} 1 & |t| < \frac{T}{2} \\ 0 & 0 \end{cases}$$

has a fourier transform $P(f) \propto \text{sinc}(\pi fT)$, where $f_0 = \frac{1}{T}$ is the symbol rate.

4. Which format can transmit the most number of bits per unit frequency?
5. Suppose that you are using a carrier of 1 GHz.
 - a) What is the corresponding wavelength λ ?
 - b) How far to you have to move the antenna to the phase by 180° ?
 - c) How many cycles per second (i.e. cycles/sec is equivalent to Hz) of the phase change occur if you are moving at:
 - i. 1 m/s (Walking)
 - ii. 30 m/s (In a car)
 - iii. 250 m/s (In a plane)

Hint: You may find the doppler formula useful:

$$\Delta f = \left(\frac{v}{c}\right) f_0$$

2 Live Signals

Since we are in the basement in EBU1, it might be difficult to capture any live signals over the air, because of that, this part of the lab will be completely optional. If you are able to find any live signals you may submit a screenshot of their spectrum for extra credit points. In this part of the lab, we will use the “*USRP_Spectral Monitoring*” VI, which is a “soft” spectrum analyzer that can display up to 32 MHz wide section of spectrum anywhere from 70 MHz to 6 GHz.

1. Open “*USRP_Spectrum Monitoring*” VI, which is a “soft” spectrum analyzer.
 - a) Some Useful Parameters for *USRP_Spectral Monitoring*:
 - i. Resolution Bandwidth $RBW = \frac{1}{T_{acq}}$ (Hz), where T_{acq} is the acquisition time or duration (in seconds).
 - ii. Frequency Span $F_{span} = f_s$, where f_s will be the IQ or sampling rate of the USRP.
 - iii. Cursor Bandwidth ΔF = the distance between the two yellow cursors of the FFT spectrum graph.
 - iv. Power in Band P = the average power within the cursor bandwidth ΔF .

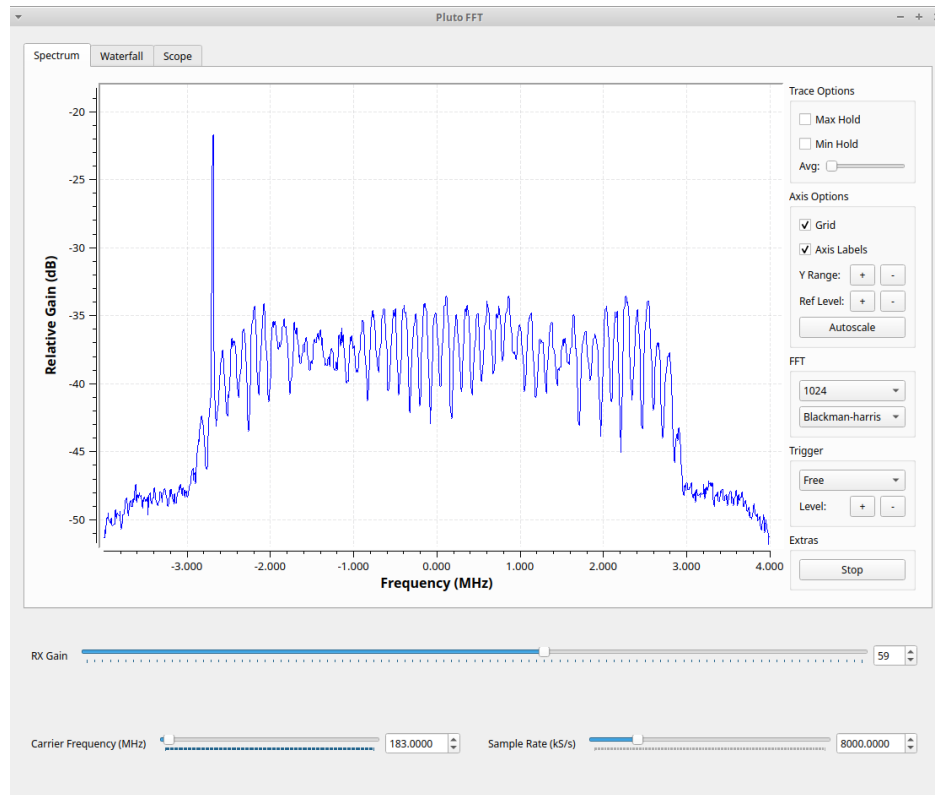


Figure 1: Front panel of the soft spectrum analyzer showing the measured spectrum of KFMB TV signal showing the carrier frequency at one edge of the passband spectrum.

2.1 Commerical TV

Commercial TV is in several bands [bands](#). In San Diego both Channel 8 (KFMB-CBS) and Channel 10 (KGTU-ABC) are in the upper VHF band with about a 6 MHz bandwidth. Channel 8 is centered at 183 MHz and Channel 10 is centered at 193 MHz. Other channels are in the UHF band with Channel 69 centered at 803 MHz. Below is the spectrum of Channel 8 taken from the University City area. Your results will be different depending on your location.

1. Try and find a digital TV channel and take a screen shot of that channel. Compare the measured carrier frequency to the table published in the Web site given earlier in this section.

2.2 Commercial FM and digital radio

The commercial FM band lies in the range of 87.5-108 MHz. Several radio channels are now broadcasting a digital signal in part of their allocated FM frequency band.

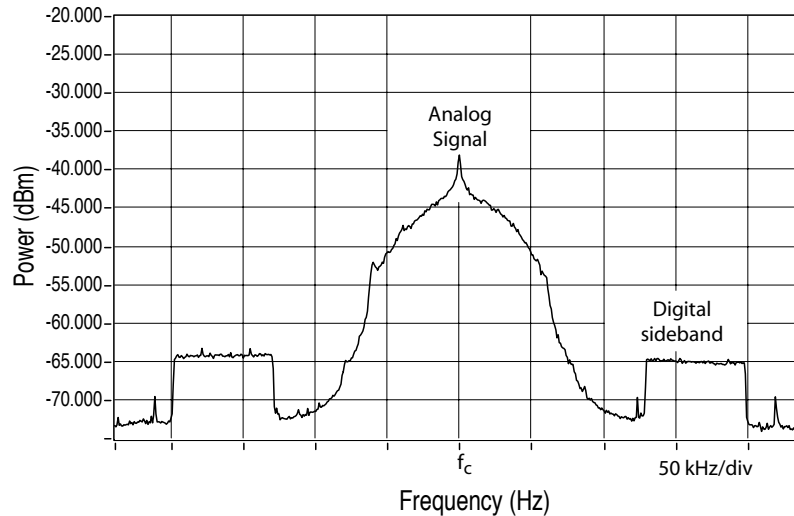


Figure 2: Measured spectrum of commercial FM signal showing both the analog signal as well as the digital sideband.

1. Scan the FM band using and try and find a channel that is broadcasting digitally based on the spectrum shown in Figure 2. Take a screen shot of that channel.

2.3 Remote Keyless System (Keyfob)

Keyless remotes contain a short-range radio transmitter, and must be within a certain range, usually 5–20 meters, of the car to work. When a button is pushed, it sends a coded signal by radio waves to a receiver unit in the car, which locks or unlocks the door. Most RKEs operate at a frequency of 315 MHz for North America-made cars and at 433.92 MHz for European, Japanese and Asian cars.

1. If you have a Keyfob with you, try tuning the center frequency in the *Spectral Monitoring VI* to the corresponding frequency for your car.
2. Next, choose an appropriate frequency span and then try to see if you can find the signal.
3. Capture any screen trace that shows the signal time-series and spectrum clearly.

2.4 PCS cellphone band (1.85-1.99 GHz)

Several cell phone carriers transmit in the PCS (personal communication services) band. There are several digital modulation formats that are used within this band including GSM (global system for mobile communications) and CDMA (code division multiple access) pioneered by San Diego-based Qualcomm. We will only qualitatively study this band.

1. Display part of the entire band using a resolution bandwidth of 100 or less. By varying the center frequency, you should see a modulated signal. If not send someone a text and see if you can see the signal. Save the spectrum of the part of the band that shows several carriers. (Note that depending on the time you are logged into the computer, you may not see any cell phone conversations that are be initiated from the CSE building.)

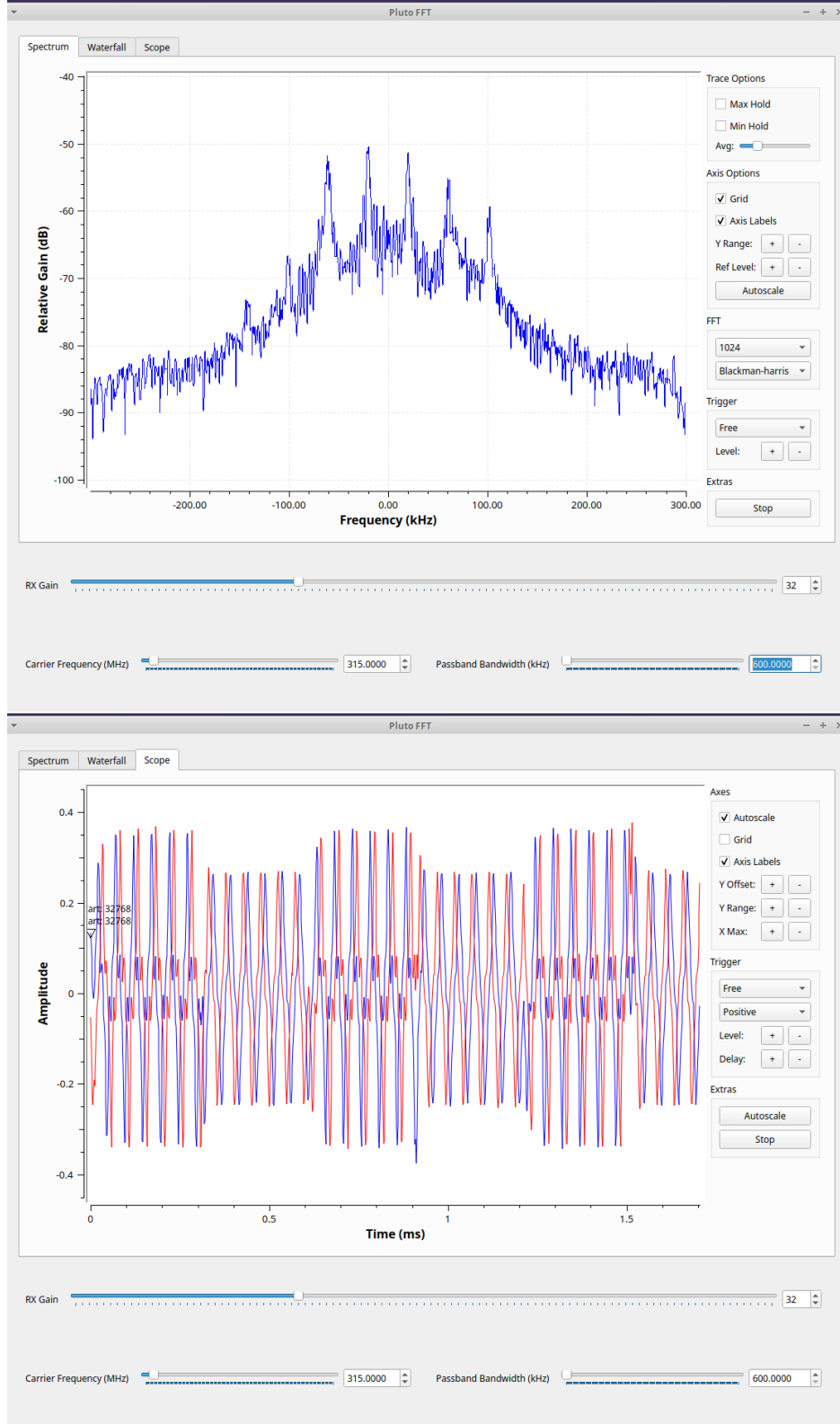


Figure 3: Top: Spectrum of remote wireless keyless entry. Bottom: Amplitude-shift-keyed time waveform using the “scope” tab on the front panel. (Note that several tries were needed to capture a clean time waveform because the signal was not synchronized.)

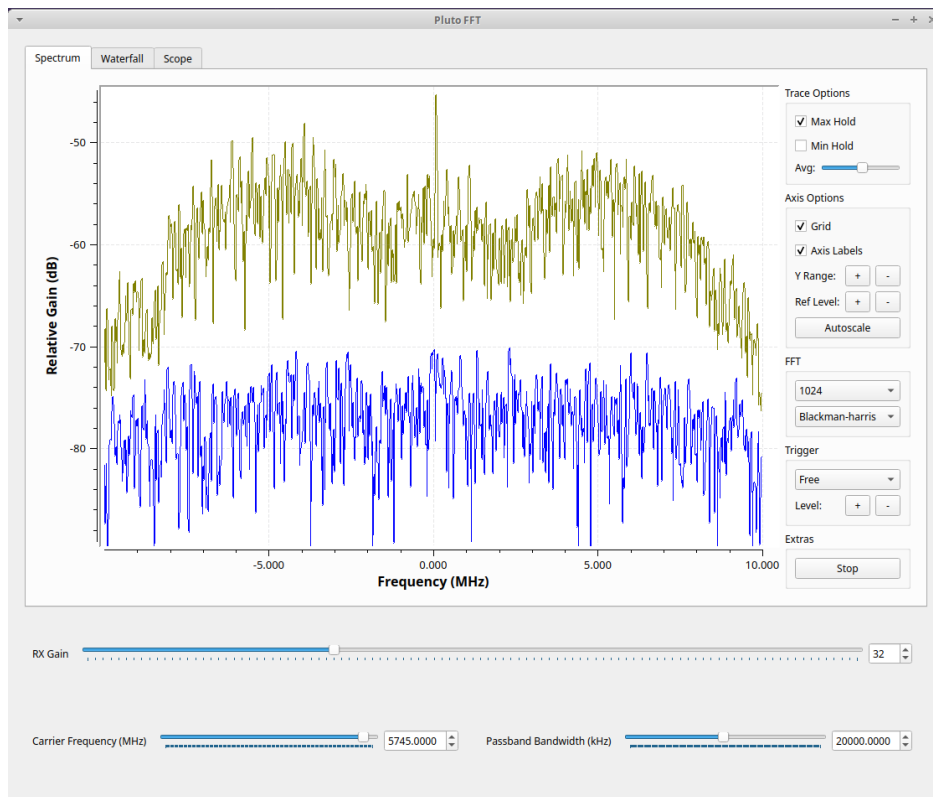


Figure 4: Blue trace: Instantaneous spectrum of Wifi Channel 192 in the 5 GHz band. Greenish trace: peak hold of trace over observation time.

2. Zoom in on a modulating band near (try 1988 MHz) and determine the bandwidth of the modulating signal.
3. Capture any screen trace that shows a signal.
4. Depending on the cell phones the other students have in lab, see if you can see the difference between a 3G signal and a 4G (LTE) signal.

2.5 WiFi Bands (2.4 and 5 GHz)

The USRP B2xx can see both the 2.4 GHz and the 5 GHz WiFi bands.

Write-Up for Part 2

1. All questions posed in the narrative.
2. The write-up for this part of the lab also consists of screen shots for the measured spectrum for any all of the communication waveforms that you could measure.